

Women and Vulnerability to HAART Non-Adherence: A Literature Review of Treatment Adherence by Gender from 2000 to 2011

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Abstract A literature review of original research articles on adherence to antiretroviral therapy (ART) in developed countries, covering January 2000 to June 2011, was conducted to determine if gender differences exist in the prevalence of nonadherence to ART. Of the 1,255 articles reviewed, only 189 included data on the proportion of the study population that was adherent and only 57 (30.2%) of these reported proportional adherence values by gender. While comparing articles was challenging because of varied reporting strategies, women generally exhibit poorer adherence than men. Thirty of the 44 articles (68.2%) that reported comparative data on adherence by gender found women to be less adherent than men. Ten articles (17.5%) reported significant differences in proportional adherence by gender,

nine of which showed women to be less adherent than men. These findings suggest that in multiple studies from developed countries, female gender often predicts lower adherence. The unique circumstances of HIV-positive women require specialized care to increase adherence to ART.

Keywords HIV · Antiretroviral therapy · Gender · Adherence · Patient compliance

Introduction

Adherence to antiretroviral therapy is an essential component of virologic suppression of HIV [1, 2]. In the post-HAART (highly active antiretroviral therapy) era (2000), adherence is considered more potent and manageable due to drug regimens with more convenient dosing schedules and improved side-effect profiles. However, subpopulations of individuals, including injection drug users (IDUs) not using methadone [3, 4], individuals using crack or cocaine, and problematically using alcohol [5–8], individuals with depression [9, 10–15], and women [7, 8, 16, 17•], have been identified as less likely to maintain an adequate rate of adherence to maintain virological suppression. Among those with poor adherence, factors such as perceived stigma [18, 19], stress [20, 21], trauma [12, 22], and lack of social support [7, 15, 21, 23] have been shown to have a negative impact on adherence.

Women have been shown to be at risk for several categories of incomplete antiretroviral therapy (ART adherence). It has been found that women have reduced uptake of ART [24, 25], higher incidence of treatment interruption [26, 27], treatment discontinuation [28], and viral rebound after initial suppression than men [8, 29, 30]. Mediating factors that may influence these findings include increased development of

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depressive symptoms in women [9•, 31, 32], reduced disclosure of serostatus [33], and support from interpersonal relationships [21]. Additionally, women are more likely than men to be excluded and/or lost to follow-up, potentially resulting in a bias that omits the most vulnerable, and potentially least adherent, women [11, 34–36]. Despite this, there is still contradicting evidence including studies that find higher adherence in women [37] or no statistical difference between genders [12, 38–42].

The purpose of this study is to review and compare ART adherence among women and men in developed countries in the modern HAART era (2000–present). In addition, factors associated with differential adherence are noted. With growing numbers of women accessing treatment and countries planning ARV rollout programs, it is important to assess the use of ART for both treatment and prevention [43].

Definition of Adherence

Multiple definitions of adherence were found in this review. The proportion of adherent individuals, rather than mean adherence per study, was examined and compared, since the authors were interested in the proportions of individuals reaching various levels of adherence. While the specific level of adherence required to reduce mortality, viral rebound, and drug resistance varies by medication, higher levels of adherence are associated with better outcomes [1, 2, 29]. Complete adherence can also include dietary compliance and appropriate timing of the dosages [44]. For the purpose of this review, articles using adherence cutoffs of 80%–100% are considered, with the adherence cutoff noted. The varying definitions of adherence used in each article add error to the overall proportion of individuals found to be adherent in this in-depth review of the literature. For example, some studies simply assessed whether or not medication was taken, while others required that the individual take medications within a specific time frame with the appropriate dietary requirements. Additionally, the method used to assess adherence can affect the resulting adherence values [44], as well as the length of time during which adherence is assessed. However, for the sake of brevity and due to the relatively small number of articles that provide proportional adherence data by gender, the articles were not subdivided by categories regarding a definition of adherence or length of time observed.

Methods

A literature review was conducted to assess adherence among men and women in developed countries during the

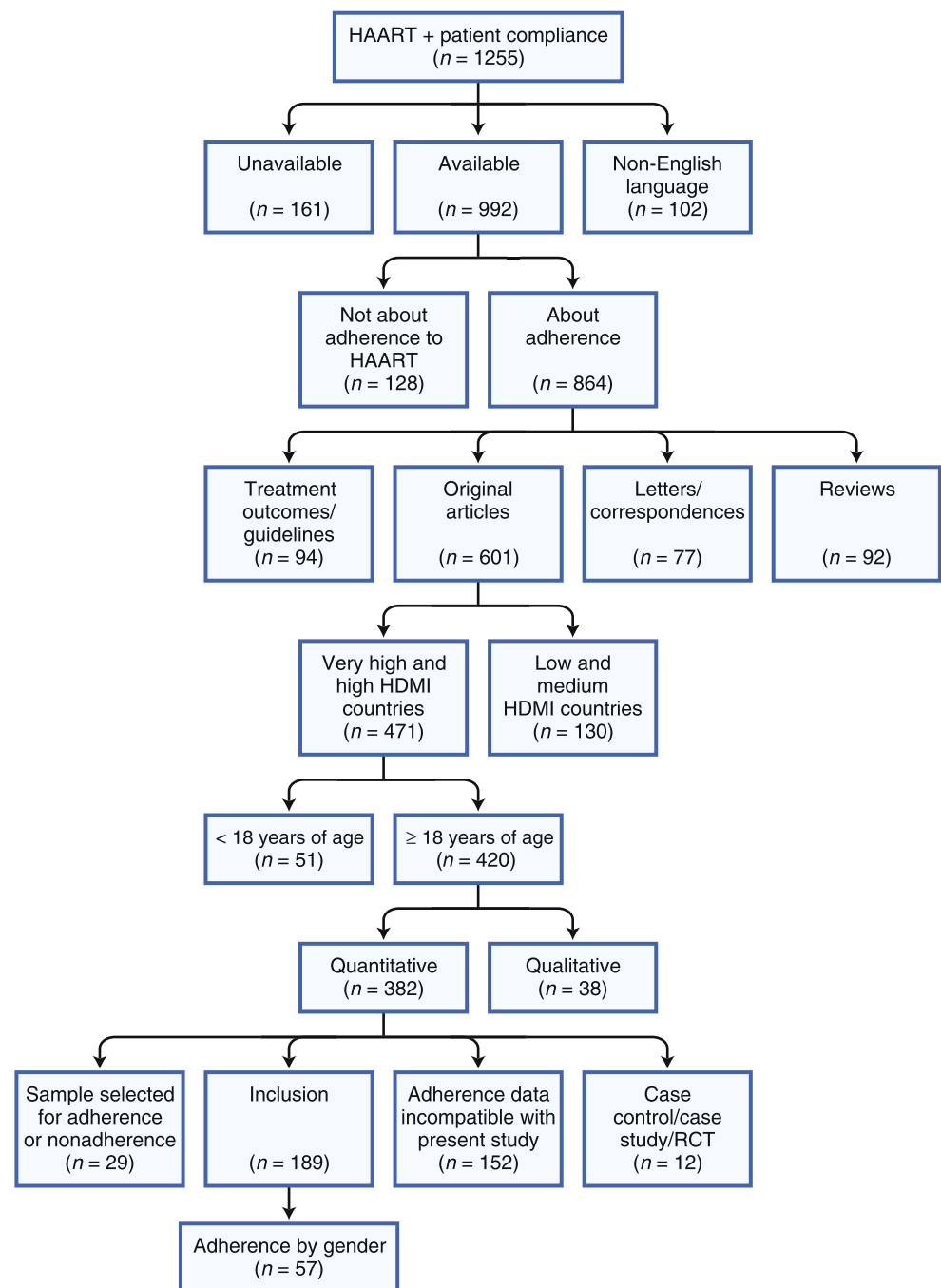
post-HAART era. The literature search was conducted using PubMed with the search terms “HAART” and “patient compliance.” The medical subject heading (MeSH) “HAART” includes antiretroviral therapy, and “patient compliance” includes both adherence and non-adherence. The review was restricted to articles from 2000 to June 2011, and includes studies examining adherence to HAART and ART. The search was purposely kept broad, so as not to exclude relevant articles. The bibliographies of identified studies were not searched for additional relevant studies. Articles were accessed via the Simon Fraser University and University of British Columbia electronic libraries.

The literature search yielded 1,255 articles, of which 1,066 did not meet the inclusion criteria for one of the following reasons (Fig. 1): the article was unavailable; published in a language other than English; described a case-control, case study, or incompatible randomized controlled trial; was in the form of a letter, correspondence, review article; described a treatment outcome or guideline without original research; or the article was not about adherence. Articles were also excluded if they presented data from Medium or Low (<0.670) Human Development Index (HDI) countries [45], included individuals under the age of 18 years, or if the study sample was purposefully selected by virtue of their adherence or non-adherence. As there are multiple definitions and expressions of adherence, articles that did not express adherence in a manner that was compatible to the definition used in this review were excluded. Qualitative articles were excluded, as they did not contribute quantitative data for analysis; however, they are referred to in order to better understand the context of adherence behavior among a diverse group of women. Articles presenting proportional adherence data by gender were included in the final analysis.

Proportional adherence data from included articles were used to calculate average proportional adherence per adherence cutoff by gender. This was also further subdivided by method of assessing adherence, including self-reported, pharmacy refill data, or other objective data, which included medication event monitoring systems (MEMS) caps and pill counts.

IDU [3, 4] and ethnicity [5, 9•, 42] were considered potential confounding variables on ART adherence by gender. In order to test whether IDU had a significant impact on adherence by gender, Fisher’s Exact Test (two-tailed) was used. The proportion of IDUs were compared between studies with significant and non-significant differences in adherence between genders. Similarly, Fisher’s Exact Test (two-tailed) was used to examine whether there was a significant difference in ethnicity (Caucasian vs non-Caucasian) between studies with significant and non-significant differences in adherence between genders.

Fig. 1 Inclusion criteria for ART proportional adherence from January 2000 to June 2011



Results

Of the initial 1,255 articles reviewed, 189 presented information on the proportion of individuals adherent to antiretroviral therapy (Fig. 1). Of these, only 57 (30.2%) reported adherence by gender. Five of the 57 studies (8.8%) lacked adherence cutoff values, and 13 (22.8%) presented data on only male or female gender and, thus, did not allow for comparison between men and women.

Table 1 compares the 57 articles that provided the proportion adherent by gender. Adherence cutoffs are

grouped together, and the means of assessing adherence and the factors associated with non-adherence are noted within the table, but not used in further analysis. Forty-four articles presented comparative adherence data on men and women, without excluding studies that did not present adherence cutoff values. Thirty of the 44 articles (68.2%) that compared proportional adherence by gender reported lower adherence among women than men. Ten of the 44 studies (22.7%) reported significant differences in adherence between genders [8, 16, 31, 37, 46, 65, 70, 73, 78, 84]. Of these, nine (90%) [8, 16, 31, 46, 65, 70, 73, 78, 84]

Table 1 Reviewed articles that included data on proportional adherence by gender

Year	Authors	Country	HDI ^a	Adherence cutoff	Adherence measure	Adherent women, <i>n</i> (%)	Adherent men, <i>n</i> (%)	Correlates of nonadherence
2011	Kyser et al. [46]	USA	0.902	100%	Self report	92 (78.0%) ^b	351 (85.6%)	Black race, unemployed and looking for work, HIV diagnosis ≥ 5 years ago, drinking 3 or more drinks per day, not engaging in aerobic activity within last 30 days
2010	Sherr et al. [42]	UK	0.849	100%	Self report	60 (59.4%)	212 (56.1%)	Young age, black ethnicity, immigrant status, greater psychological symptom burden
2009	Kumar and Encinosa [10]	USA	0.902	100%	Self report	143 (48.6%)	547 (60.9%)	Untreated high depressive symptoms
2009	Royal et al. [47]	USA	0.902	100%	Self report	80 (81.6%)	200 (77.5%)	Young age, no health insurance, drug use, low mental health, depression, stress, negative attitudes toward treatment
2008	Miguez-Burbano et al. [48]	USA	0.902	100%	Self report	—	41 (30.4%)	Gastrointestinal adverse effects, body changes, sexual dysfunction, low HAART efficacy expectations, lack of partner support, depression
2008	Sherr et al. [49]	UK	0.849	100%	Self report	45 (41.3%)	168 (43.4%)	Young age, non-white ethnicity, low quality of life, psychological symptoms, global distress symptoms
2007	Beach et al. [50]	USA	0.902	100%	Self report	(76.1%)	(80.7%)	No involvement in health decision making
2006	Peretti-Watel et al. [51]	France	0.872	100%	Self report	393 (61.1%)	1086 (59.0%)	Alcohol abuse or heroin use with nicotine dependence, cannabis use, or amyl nitrate use
2006	Peretti-Watel et al. [52]	France	0.872	100%	Self report	225 (55.1%)	824 (58.8%)	Unsafe sex in women
2006	Rintamaki et al. [19]	USA	0.902	100%	Self report	28 (68.3%)	114 (69.9%)	High stigma
2005	Phillips et al. [13]	USA	0.902	100%	Self report	52 (30%)	—	Sleep disturbance, depression
2004	Ammassari et al. [14]	Italy	0.854	100%	Self report	33 (68.8%)	61 (70.1)	Depressive symptoms, IDU, sexual dysfunction
2004	Cook et al. [53]	USA	0.902	100%	Self report	311 (37%)	—	—
2004	Deloria-Knoll et al. [54]	USA	0.902	100%	Self report	21 (60.0%)	150 (68.2%)	Young age, unemployment, alcohol use, depression
2004	Kleeberger et al. [55]	USA	0.902	100%	Self report	—	836 (74.1%)	—
2004	Palepu et al. [56]	Canada	0.888	100%	Pharmacy data	132 (43.1%)	912 (63.3%)	Female, young age, low physician experience, history of incarceration, history of IDU, low median baseline HIV-1 RNA, high median baseline CD4 cell count
2004	Sharpe et al. [57]	USA	0.902	100%	Self report	534 (68.1%)	—	Illicit drug use
2004	Wood et al. [21]	USA	0.902	100%	Self report	30 (83.4%)	—	Interpersonal relationships, level of distress
2003	Fong et al. [58]	Hong Kong	0.862	100%	Self report	17 (89.5%)	113 (79.6%)	Gastrointestinal adverse effects
2002	Spire et al. [59]	France	0.872	100%	Self report	70 (72.2%)	348 (78.2%)	Young age, poor housing conditions, lack of social support, previous problems with adherence, side effects, perception of one's health, belief of HAART effectiveness and

Table 1 (continued)

Year	Authors	Country	HDI ^a	Adherence cutoff	Adherence measure	Adherent women, <i>n</i> (%)	Adherent men, <i>n</i> (%)	Correlates of nonadherence
2002	Wagner et al. [60]	USA	0.902	100%	Self report	1 (14.3%)	20 (60.6%)	toxicity, increased alcohol and tobacco use, contact with non-regular physicians
2001	Brigido et al. [61]	Brazil	0.699	100%	Self report	26 (46.4%)	49 (38.9%)	Lower formal education, low perceived treatment efficacy, low treatment knowledge
2001	Duran et al. [62]	France	0.872	100%	Self report	49 (73.1%)	203 (75.5%)	Medication intolerance, alcohol use, misunderstanding prescription
2006	Gross et al. [63]	Canada	0.888	>95%	Pharmacy data	213 (82.6%)	1166 (84.7%)	High number of symptoms at baseline
2009	Applebaum et al. [32]	USA	0.902	≥95%	Self report	15 (68.2%)	32 (71.1%)	—
2009	Plankey et al. [64]	USA	0.902	≥95%	Self report	1270 (76.0%)	—	Side effects and alcohol use, particularly for women
2008	Fielden et al. [16]	Canada	0.888	≥95%	Pharmacy data	110 (40.7%) ^b	833 (62.4%)	Self-perception of abdominal fat gain, young age, black ethnicity, history of IDU, higher HIV RNA at previous visit, alcohol use
2008	Fumaz et al. [65]	Spain	0.863	≥95%	Self report/ plasma concentration	13 (46.4%) ^b	41 (78.8%)	Female, young age, starting HAART prior to AIDS diagnosis, lower CD4 cell counts, less experienced physician, hospitalization in 6 months prior to baseline, IDU
2008	Kapadia et al. [66]	USA	0.902	≥95%	Self report	421 (73.5%)	—	Female, current substance use, hepatitis C coinfection
2008	Merenstein et al. [67]	USA	0.902	≥95%	Electronic device	684 (50.1%)	—	Non-participation in drug treatment programs
2008	Ubbiali et al. [17••]	Italy	0.854	≥95%	Self report	66 (48.5%)	202 (59.1%)	Higher number of children, black ethnicity, drug use
2007	Graham et al. [68]	USA	0.902	≥95%	Pharmacy data	12 (54.5%)	33 (50.8%)	Female, greater need for approval and preoccupation with relationships
2007	Vaz et al. [69]	Brazil	0.699	≥95%	Plasma concentration	14 (17.7%)	—	Low literacy
2006	Braitstein et al. [70]	Canada	0.888	≥95%	Pharmacy data	61 (35.5%) ^b	614 (60.6%)	Not being pregnant, ≥6 pills per day, age younger than 29
2006	Palepu et al. [4]	Canada	0.888	≥95%	Pharmacy data	48 (40.7)	81 (50.6%)	Hepatitis C coinfection, female, young age, IDU, not having AIDS diagnosis at baseline
2006	Sodergard et al. [71]	Sweden	0.885	≥95%	—	—	354 (59.2%)	Female, heroin use, not using methadone, young age, baseline CD4 cell count
2005	Anastos et al. [72]	USA	0.902	≥95%	—	222 (74.7%)	—	Drug and alcohol use, young age, longer time on therapy
2005	Bonolo et al. [73]	Brazil	0.699	≥95%	Self report	57 (53.3%) ^b	136 (68.3%)	—
2005	Halkitis et al. [74]	USA	0.902	≥95%	Electronic device	—	182 (60.7%)	Female, unemployment, alcohol use, 3 or more adverse reactions, number of pills per day, switch in ART regimen, longer time between result and prescription
2004	Kerr et al. [75]	Canada	0.888	≥95%	Pharmacy data	18 (36.7%)	19 (32.2%)	Access to health care, cocaine use
								Low efficacy expectations, self-regulatory efficacy, outcome expectation

Table 1 (continued)

Year	Authors	Country	HDI ^a	Adherence cutoff	Adherence measure	Adherent women, <i>n</i> (%)	Adherent men, <i>n</i> (%)	Correlates of nonadherence
2004	Kuyper et al. [8]	Canada	0.888	≥95%	Pharmacy data	73 (57.9%) ^b	643 (76.2%)	Female, IDU, young age
2004	Mohammed et al. [76]	USA	0.902	≥95%	Self report	41 (64.1%)	100 (66.2%)	Alcohol use
2004	Palepu et al. [77]	USA	0.902	≥95%	Self report	33 (82.5%)	113 (73.4%)	Alcohol and drug use
2003	Delgado et al. [78]	Canada	0.888	≥95%	Pharmacy data	40 (33.3%) ^b	455 (59.4%)	Female, IDU, young age, higher CD4 cell count, pill burden, not diagnosed with AIDS
2003	Turner et al. [31]	USA	0.902	≥95%	Pharmacy data	334 (18%) ^b	812 (25%)	Female, depression
2006	Parruti et al. [79]	Italy	0.854	≥90%	Self report/ pharmacy data	50 (89.3%)	101 (87.8%)	Homeless, drug-related toxicity
2002	Garcia de Olalla et al. [80]	Spain	0.863	≥90%	Self report/ pharmacy data	237 (66.6%)	594 (68.8%)	—
2002	Witteveen and van Ameijden [81]	Netherlands	0.890	≥90%	Self report	7 (63.6%)	8 (50.0%)	Medication side effects, poor acceptance of serostatus, doubts of medication effectiveness, irregular use of hard drugs, depression, poor knowledge of HIV, poor physical health, homelessness, low income, negative attitude from health care practitioners, little/no social support
2001	Martin et al. [82]	Spain	0.863	≥90%	Pharmacy data	38 (54.3%)	75 (43.6%)	—
2000	Roca et al. [37]	Spain	0.863	>80%	Self report	12 (66.7%)	18 (38.3%) ^b	Male, high baseline HIV RNA
2000	Moatti et al. [83]	France	0.872	≥80%	Self report	33 (63.5%)	74 (66.1%)	Young age, alcohol use, frequency of negative life-events during the prior 6 months, active drug use
2009	Duggan et al. [84]	USA	0.902	—	Self report/ clinical data	26 (63.4%) ^b	70 (79.5%)	Belief in effectiveness of therapy, difficulty taking medication, thinking of HIV as “punishment”
2008	Cardarelli et al. [85]	USA	0.902	—	Self report	5 (19.2%)	23 (30.7%)	Low general health
2008	Shin et al. [23]	Peru	0.723	—	Self report	7 (58.3%)	20 (64.5%)	Low social support
2006	Garcia et al. [86]	Brazil	0.699	—	Self report	57 (65.5%)	59 (62.1%)	Longer duration of therapy, low belief in the value of therapy, therapy complexity, being away from home
2006	Moralejo et al. [87]	Spain	0.863	—	Self report/ pharmacy data	29 (61.7%)	67 (69.8%)	Unemployment, housing situation, degree of treatment acceptance
2004	Russell et al. [88]	USA	0.902	—	Provider assessed	4 (80.0%)	104 (83.2%)	Low provider satisfaction, unemployment

^a Human development index [46]^b Significant differences between genders ($P \leq 0.05$)

showed significantly lower ($P \leq 0.05$) adherence among women than men. Four of the articles finding a significant difference were based on Canadian study samples, three were based on American samples, and the remainder were based on samples from South America and Europe. The

single study that yielded significantly lower adherence among men [37] was conducted in Spain and used an adherence cutoff of 80%. It should be noted that the majority of articles did not specify whether there was a significant relationship between gender and adherence, as

adherence by gender was not the main focus and/or because there was no significant relationship to report.

Table 2 lists mean proportional adherence by gender, splitting by adherence cutoff and the methods by which adherence was assessed. When multiple means of assessing adherence were used, such as self-report in conjunction with pharmacy refill data, the data were categorized as other objective methods. Using cutoffs of 100% and $\geq 95\%$, women had lower mean proportional adherence than men in all adherence measurement categories. Using cutoffs of $\geq 90\%$ and $\geq 80\%$, women were found to have a higher mean proportional adherence than men in almost all measurement categories.

When examining all studies which reported a *P* value regarding differences in adherence by gender and the proportion of IDUs in the study, a significantly higher proportion ($P < 0.0001$) of IDUs were found in studies that yielded no significant difference in proportional adherence between men and women. However, this included two studies [4, 75] that were restricted to IDUs, neither of which showed significant differences in adherence between men and women. When these two studies were excluded, there was no significant difference ($P = 0.4553$) between the proportion of IDUs in studies that had a significant difference in adherence between genders and those that did not.

The proportion of Caucasian, compared to non-Caucasian, respondents in the studies was significantly different ($P < 0.0001$). Studies with no significant differences in adherence between genders were found to have a higher proportion of Caucasian respondents than studies with significant difference between genders. Results from Fisher's Exact Test for both IDUs and ethnicity should be interpreted with caution, due to the small sample sizes used in both tests.

Discussion

The literature suggests that women have lower adherence than men. Lower adherence in women may be due to multiple conditions more common in women. Reduced

adherence was found to be associated with depression [13, 48], lack of supportive interpersonal relationships [48, 58], young age [64, 69, 71], and drug [57, 67, 71] and alcohol use [64, 71]. Additionally, non-adherence in women specifically was associated with black ethnicity [64, 67], non-pregnancy, ART of six or more pills per day [69], women with a higher number of children [67], self-perception of abdominal fat gain [64], sleep disturbances [13], and increased levels of distress [58].

Several differences between genders were observed in comparative studies. Non-adherent women place greater weight on their relationships with others, having a greater need for approval and a preoccupation with relationships, whereas non-adherent men focused less on relationships [17•]. Among HIV-positive individuals on methadone maintenance, physical and emotional functioning quality of life was lower in women than in men [89]. Medication side effects and alcohol use were associated with non-adherence, particularly among women [32]. One study [52] found non-adherence to be correlated with unsafe sex in heterosexual women, when compared to heterosexual and homosexual men. Additionally, greater adherence was observed among gay men, followed by heterosexual men, and the poorest adherence was observed among heterosexual women [60].

These differences highlight that men's and women's experiences with HIV and adherence to ART are different, and therefore gender-specific adherence support may be necessary for optimal adherence. Emotional distress, stigmatization, and depression may be intensified among women, particularly if they do place greater emphasis on their relationships with others and lack strong social support. This can lead to non-disclosure out of fear, self-imposed isolation, and deepening depression. In concert, these psychosocial variables can lead to reduced adherence among women.

This is not to suggest, however, that women ought to be viewed as a homogenous group that tends to exhibit lower adherence than men. Women can, and do, adhere well to ART, as shown by the great variability in proportional adherence in this review. However, it is potentially the interaction of gender with other factors, such as poverty, homelessness, sexual

Table 2 Mean proportional adherence by gender per adherence cutoff and means of assessing adherence

	Gender	100%	$\geq 95\%$	$\geq 90\%$	$\geq 80\%$
All measures% (range)	Female	59.8 (14.3–89.5)	52.7 (17.7–82.6)	68.5 (54.3–89.3)	65.1 (63.5–66.7)
	Male	64.5 (30.4–85.6)	61.3 (25–84.7)	62.6 (43.6–87.8)	52.2 (38.3–66.1)
Self-reported% (range)	Female	60.6 (14.3–89.5)	64.1 (46.4–82.5)	63.6 (–)	—
	Male	64.6 (30.4–85.6)	69.5 (59.1–78.8)	50.0 (–)	—
Pharmacy refill data% (range)	Female	43.1 (–)	44.4 (18–82.6)	54.3 (–)	—
	Male	63.3 (–)	55.8 (25–84.7)	43.6 (–)	—
Other objective measures% (range)	Female	—	33.9 (17.7–50.1)	78.0 (66.6–89.3)	—
	Male	—	60.7 (–)	78.3 (68.8–87.8)	—

orientation, drug use, marital and relationship status, and ethnicity, that further modifies adherence. With this in mind, a gendered, intersectional approach to adherence would be beneficial to identify how to best support adherence.

Future research should not neglect gender when assessing adherence, and factors associated with low adherence in both men and women should be identified. Interventions that provide social support with an understanding of the role that personal relationships can have may improve women's adherence. Studies on interventions, however, should note if there is a difference in adherence improvement by gender and/or other factors.

Limitations

There are several limitations that should be noted in this study. As this was a literature review using one search engine, it is likely that some studies about adherence have been omitted. However, the authors used broad search terms and examined a relatively large sample of articles in order to minimize error due to omission. Secondly, statistics were drawn from articles based on original research. The authors found articles that used the same cohort, however, adherence data reported in articles sharing the same cohort were not identical, as there may have been different inclusion and exclusion criteria.

The varying definitions of adherence and cutoff values were problematic for the comparison of adherence values. As there were a limited number of articles comparing adherence data by gender, proportional adherence was compared by adherence cutoff, and not by a definition of adherence that includes dietary and schedule compliance. The majority of studies defined adherence as taking all pills, and did not define adherence by dietary and scheduling compliance. As such, the adherence values in this report may be higher than they would be if a strict definition of adherence including dietary and scheduling compliance were used.

With a relatively small number of studies that report proportional adherence by gender, the authors could not individually categorize data by the means which adherence was assessed, such as self-report, pharmacy refill data, MEMS caps, pill counts, physician assessment, and plasma concentrations of medications. The authors chose to assess mean adherence by combined assessment measures, self-report, pharmacy refill data, and other objective methods. Self-report was the most common means of assessing adherence, used in 130 of the 189 studies (68.8%) reporting proportional adherence. Pharmacy refill data was used in 34 of 189 studies (18.0%). Other objective methods were used in 25 of 189 studies (13.2%). Adherence data by gender was not further stratified by duration of observation due to a high level

of variability and a relatively small number of articles with adherence data by gender. The authors note that this would influence the proportion of individuals reported to be adherent, but the high level of variability and a relatively small sample of articles precludes subdividing the analysis.

It should also be noted that very few studies provided data on the significance of the relationship between adherence and gender, and the proportion of IDUs within the study. As such, only nine studies that showed a significant difference between men and women, and five studies which noted an insignificant difference between men's and women's adherence, were used in Fisher's Exact Test.

The high and very high Human Development Index scores (≥ 0.670) [45] were used as an exclusion criteria in order to attempt to homogenize the study populations reviewed as much as possible. It is unlikely that this measure would completely reduce regional biases on adherence. This also makes it difficult to ascertain social and cultural constructs which shape adherence. Multiple aspects of one's environment will produce barriers and facilitators to medication adherence. Additionally, there is variable access to ART between countries, as not all countries offer coverage of ART. More qualitative studies are needed to identify the barriers and facilitators to adherence within populations [89–92], as well as between genders [93].

The authors were unable to conduct analyses on the effects of confounding variables, such as IDU, income, education level, and homelessness or unstable housing, since these variables were seldom compared with adherence and gender together. Similarly, the role of depressive symptoms and ART side effects could not be assessed as few studies examined these with adherence and gender.

Another limitation is that women made up a comparatively small proportion of study samples, and are disproportionately excluded from some adherence studies or are more likely to be lost to follow-up [11, 34–36]. If women are indeed more prone to rely on social support and affirmation from social relationships, as is suggested by Ubbiali et al. [17•], it is possible that self-reported values of adherence are inflated due to social desirability bias. Additionally, it is possible that the most marginalized women, and those most at risk of suboptimal adherence, are being excluded from studies or reporting falsely high levels of adherence. This may result in an adherence value that is biased toward greater proportional adherence. Focus should be placed on inclusive sampling and greater outreach to avoid loss to follow-up among vulnerable groups.

Conclusions

This in-depth review of proportional adherence to ART in high-income countries found that women are generally less

adherent than men. Only 57 of 189 articles (30.2%) with adherence data reported adherence by gender. Of the 44 articles that compared adherence by gender, women were reported as less adherent than men in 30 studies (68.2%), although the associations were not significant in every study. Of the 10 articles that provided data on the significance of proportional adherence by gender, nine (90.0%) exhibited significantly lower adherence among women than men. This may be an artifact of women's disproportionate inclusion in the IDU risk category in some settings, however, women who inject drugs still had significantly poorer adherence to therapy than men who inject drugs [8]. This may reflect women's greater focus on relationships and the detrimental effects of stigmatization [17••].

More research into a gendered perspective on adherence is needed to determine if injection drug use is a modifying factor in women's suboptimal adherence, or if women are less adherent due to different barriers than those experienced by men. As suggested by Ubbiali et al. [17••], different factors facilitate or inhibit ART adherence for women and men. Variability between studies may be a result of social, psychological, and economic characteristics and may be specific to a given population or environment. Women's experiences with HIV-related depression, stress, and stigmatization [9•, 11, 31, 32, 52] and a greater focus on social relationships likely affects their levels of adherence. Although differences in proportional adherence are inconclusive in this review, gender is an important component of one's identity, influencing one's concept of HIV and means by which it is managed. Future adherence studies should aim to disaggregate adherence data by gender, and attempt to control for potentially confounding factors such as injection drug use, poverty, relationship status, ethnicity, and sexual orientation. Qualitative studies are also needed to understand the subtleties of men and women's experience with ART adherence. A gendered-perspective of adherence can help shape more appropriate and targeted support for individuals accessing ART.

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